

This assignment is Questions 1 and 3 of the Autumn Repeat Exam of the 2015-2016 academic year. The exam paper is on the module webpage. I don't repeat all questions from the exam below.

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1. Let $|\phi_n\rangle$ be the orthonormalized energy eigenstate of a particle of mass m in an infinite square well of width a , with corresponding energy eigenvalue

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \quad n = 1, 2, 3, \dots$$

The particle is prepared to be in the state

$$|\psi\rangle = \sum_{n=1}^{\infty} \frac{\alpha}{n^2} |\phi_n\rangle$$

(a) –

Hint/Solution $\rightarrow$

Since the $|\phi_n\rangle$ are orthonormalized,

$$\langle \phi_m | \phi_n \rangle = \delta_{mn}$$

The norm of the state $|\psi\rangle = \sum \frac{\alpha}{n^2} |\phi_n\rangle$ is therefore

$$\begin{aligned} \langle \psi | \psi \rangle &= \left(\sum_{n=1}^{\infty} \frac{\alpha^*}{n^2} \langle \phi_n | \right) \left(\sum_{n=1}^{\infty} \frac{\alpha}{n^2} |\phi_n\rangle \right) \\ &= \left(\sum_{l=1}^{\infty} \frac{\alpha^*}{l^2} \langle \phi_l | \right) \left(\sum_{n=1}^{\infty} \frac{\alpha}{n^2} |\phi_n\rangle \right) \quad \left\{ \begin{array}{l} \text{Changed a dummy variable} \\ \text{to avoid confusion} \end{array} \right. \\ &= \sum_{l=1}^{\infty} \sum_{n=1}^{\infty} \frac{\alpha^*}{l^2} \frac{\alpha}{n^2} \langle \phi_l | \phi_n \rangle = |\alpha|^2 \sum_{l=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{l^2} \frac{1}{n^2} \delta_{ln} \end{aligned}$$

If we perform the summation over n , only the term $n = l$ survives, by definition of the Kronecker delta δ_{ln} . Hence

$$\langle \psi | \psi \rangle = |\alpha|^2 \sum_{l=1}^{\infty} \frac{1}{l^2} \frac{1}{l^2} = |\alpha|^2 \sum_{l=1}^{\infty} \frac{1}{l^4} = |\alpha|^2 \frac{\pi^4}{90}$$

The result $\pi^4/90$ for the infinite series is given in the exam paper.

To be normalized, we require the norm to be 1; hence

$$|\alpha|^2 = \frac{90}{\pi^4} \quad \Rightarrow \quad \alpha = \frac{3\sqrt{10}}{\pi^2} \times \text{phase factor}$$

(b) —

Hint/Solution $\rightarrow$

Since the $|\phi_n\rangle$ are energy eigenstates,

$$\hat{H}|\phi_n\rangle = E_n|\phi_n\rangle \quad \langle\phi_l|\hat{H}|\phi_n\rangle = E_n\langle\phi_l|\phi_n\rangle = E_n\delta_{ln}$$

The expectation value of energy is

$$\begin{aligned} \langle\psi|\hat{H}|\psi\rangle &= \left(\sum_{n=1}^{\infty} \frac{\alpha^*}{n^2} \langle\phi_n|\right) \hat{H} \left(\sum_{n=1}^{\infty} \frac{\alpha}{n^2} |\phi_n\rangle\right) \\ &= \left(\sum_{l=1}^{\infty} \frac{\alpha^*}{l^2} \langle\phi_l|\right) \hat{H} \left(\sum_{n=1}^{\infty} \frac{\alpha}{n^2} |\phi_n\rangle\right) \quad \left\{ \begin{array}{l} \text{using different summation variables} \\ \text{to avoid confusion} \end{array} \right. \\ &= \sum_{l=1}^{\infty} \sum_{n=1}^{\infty} \frac{\alpha^*}{l^2} \frac{\alpha}{n^2} \langle\phi_l|\hat{H}|\phi_n\rangle = |\alpha|^2 \sum_{l=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{l^2} \frac{1}{n^2} E_n \delta_{ln} \\ &= |\alpha|^2 \sum_{l=1}^{\infty} \frac{E_l}{l^4} = \frac{90}{\pi^4} \sum_{l=1}^{\infty} \frac{E_l}{l^4} = \frac{90}{\pi^4} \sum_{n=1}^{\infty} \frac{E_n}{n^4} \end{aligned}$$

Using $E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2$, we obtain

$$\langle\psi|\hat{H}|\psi\rangle = \frac{45\hbar^2}{ma^2\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{45\hbar^2}{ma^2\pi^2} \frac{\pi^2}{6} = \frac{15\hbar^2}{2ma^2}$$

- (c) What is the probability that a measurement of the energy gives $2\pi^2\hbar^2/ma^2$, and if this energy is found, what state is the particle in immediately afterwards?

Hint/Solution $\rightarrow$

A measurement of energy results necessarily in one of the eigenvalues, E_n . The probability of finding the energy E_n , i.e., the probability of finding the system in the state $|\phi_n\rangle$, is $|\langle\phi_n|\psi\rangle|^2$. Equivalently, if the state $|\psi\rangle$ is expanded in the eigenstates $|\phi_n\rangle$, the mod-square of the coefficient of $|\phi_j\rangle$ gives the probability of finding the energy to be E_j . If the energy is found to be E_j , then the system will be in the state $|\phi_j\rangle$ immediately after the measurement.

A measurement of energy will give the result

$$\frac{2\pi^2\hbar^2}{ma^2} = \frac{\pi^2\hbar^2}{2ma^2}2^2 = E_2$$

with a probability $|\langle\phi_2|\psi\rangle|^2$. Noting that

$$\langle\phi_2|\psi\rangle = \langle\phi_2|\left(\sum_{n=1}^{\infty}\frac{\alpha}{n^2}|\phi_n\rangle\right) = \sum_{n=1}^{\infty}\frac{\alpha}{n^2}\langle\phi_2|\phi_n\rangle = \sum_{n=1}^{\infty}\frac{\alpha}{n^2}\delta_{2,n} = \frac{\alpha}{2^2}$$

we get for the probability

$$|\langle\phi_2|\psi\rangle|^2 = \frac{|\alpha|^2}{16} = \frac{1}{16}\frac{90}{\pi^4} = \frac{45}{8\pi^4} \approx 0.0577$$

If the energy is found to be E_2 , then the system will be in the state $|\phi_2\rangle$ immediately after the measurement.

3. (a) —

Hint/Solution $\rightarrow$

Since the wavefunction is composed only of the two eigenstates ϕ_2 and ϕ_3 , the only possible values of the energy are

$$E_2 = \frac{\pi^2 \hbar^2}{2ma^2} 2^2 \quad \text{and} \quad E_3 = \frac{\pi^2 \hbar^2}{2ma^2} 3^2$$

The probabilities are the mod-squared of the coefficients of the corresponding terms. The probability of finding energy E_2 is

$$\left| \frac{3i}{\sqrt{10}} \right|^2 = \frac{9}{10}$$

and the probability of finding energy E_3 is

$$\left| \frac{1}{\sqrt{10}} \right|^2 = \frac{1}{10}$$

(b) Compute $\langle \hat{x} \rangle$ and $\langle \hat{p} \rangle$ as functions of time.

Hint/Solution $\rightarrow$

Position

$$\begin{aligned}\langle \hat{x} \rangle &= \langle \psi | \hat{x} | \psi \rangle = \int_0^a dx \psi^*(t, x) \hat{x} \psi(t, x) = \int_0^a dx \psi^*(t, x) x \psi(t, x) \\ &= \int_0^a dx x |\psi(t, x)|^2 = \int_0^a dx x \left| \frac{3i}{\sqrt{10}} e^{-i\omega_2 t} u_2(x) + \frac{1}{\sqrt{10}} e^{-i\omega_3 t} u_3(x) \right|^2 \\ &= \int_0^a dx x \left(\frac{9}{10} |u_2(x)|^2 + \frac{1}{10} |u_3(x)|^2 + 2\text{Re} \left[\frac{3i}{10} e^{i(\omega_3 - \omega_2)t} u_2(x) u_3^*(x) \right] \right)\end{aligned}$$

where we have used $|a + b|^2 = |a|^2 + |b|^2 + 2\text{Re}[ab^*]$. Note that

$$\int_0^a dx x |u_2(x)|^2$$

is the expectation value of position in the eigenstate $u_2(x)$, which we expect to be $a/2$; we did a similar calculation in Problem Set 5. You can of course insert $u_2(x) = \sqrt{\frac{2}{a}} \sin(2\pi(x)/a)$ (re-)calculate the integral. The same for the expectation value of position in the eigenstate $u_3(x)$; thus

$$\int_0^a dx x |u_2(x)|^2 = \int_0^a dx x |u_3(x)|^2 = \frac{a}{2}$$

Also, $\text{Re} [ie^{i(\omega_3 - \omega_2)t}] = -\sin[(\omega_3 - \omega_2)t]$. Thus

$$\begin{aligned}\langle \hat{x} \rangle &= \frac{9}{10} \frac{a}{2} + \frac{1}{10} \frac{a}{2} - \sin[(\omega_3 - \omega_2)t] \frac{3}{5} \int_0^a dx x u_2(x) u_3^*(x) \\ &= \frac{a}{2} - A \sin[(\omega_3 - \omega_2)t]\end{aligned}$$

Thus the expectation value of position oscillates around the center of the potential well with frequency

$$\omega_3 - \omega_2 = \frac{E_3 - E_2}{\hbar} = \frac{5\pi^2 \hbar}{2ma^2}$$

We still have to calculate the amplitude of oscillation:

$$A = \frac{3}{5} \int_0^a dx x u_2(x) u_3^*(x) = \frac{3}{5} \times \frac{2}{a} \int_0^a dx x \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{3\pi x}{a}\right)$$

The integral needs some work; I won't do it out. It turns out to be $-\frac{24a^2}{25\pi^2}$, so that

$$A = -\frac{144a}{125\pi^2}$$

so that

$$\langle \hat{x} \rangle = \frac{a}{2} + \frac{144a}{125\pi^2} \sin[(\omega_3 - \omega_2)t] = \frac{a}{2} + \frac{144a}{125\pi^2} \sin\left[\frac{5\pi^2 \hbar t}{2ma^2}\right]$$

Momentum

$$\begin{aligned} \langle \hat{p} \rangle &= \langle \psi | \hat{x} | \psi \rangle = \int_0^a dx \psi^*(t, x) \hat{p} \psi(t, x) \\ &= \int_0^a dx \left[-\frac{3i}{\sqrt{10}} e^{+i\omega_2 t} u_2(x) + \frac{1}{\sqrt{10}} e^{+i\omega_3 t} u_3(x) \right] \\ &\quad \times \hat{p} \left[\frac{3i}{\sqrt{10}} e^{-i\omega_2 t} u_2(x) + \frac{1}{\sqrt{10}} e^{-i\omega_3 t} u_3(x) \right] \\ &= \frac{9}{10} \int_0^a dx u_2 \hat{p} u_2 + \frac{1}{10} \int_0^a dx u_3 \hat{p} u_3 \\ &\quad - \frac{3i}{10} e^{-i(\omega_3 - \omega_2)t} \int_0^a dx u_2 \hat{p} u_3 + \frac{3i}{10} e^{+i(\omega_3 - \omega_2)t} \int_0^a dx u_3 \hat{p} u_2 \end{aligned}$$

We have used the fact that, since $\hat{p} = -i\hbar\partial_x$, this operator leaves the time-dependent phase-factors unchanged. The space-dependence of u_2 and u_3 are left implicit in the last lines above.

The first two terms represent the expectation value of momentum in the state $u_2(x)$ and in the state $u_3(x)$. Since these are localized bound states with no net motion, the momentum expectation value is zero. This was calculated in Problem Set 5 for $u_1(x)$. Thus

$$\begin{aligned} \langle \hat{p} \rangle &= -\frac{3i}{10} e^{-i(\omega_3 - \omega_2)t} \int_0^a dx u_2 \hat{p} u_3 + \frac{3i}{10} e^{+i(\omega_3 - \omega_2)t} \int_0^a dx u_3 \hat{p} u_2 \\ &= -\frac{3\hbar}{10} e^{-iRt} \int_0^a dx u_2 \partial_x u_3 + \frac{3\hbar}{10} e^{+iRt} \int_0^a dx u_3 \partial_x u_2 \quad \left\{ \begin{array}{l} \text{introduce notation} \\ R = (\omega_2 - \omega_1) \end{array} \right. \\ &= -\frac{3\hbar}{10} \frac{2}{a} \frac{3\pi}{a} e^{-iRt} \int_0^a dx \sin\left(\frac{2\pi x}{a}\right) \cos\left(\frac{3\pi x}{a}\right) + \frac{3\hbar}{10} \frac{2}{a} \frac{2\pi}{a} e^{+iRt} \int_0^a dx \sin\left(\frac{3\pi x}{a}\right) \cos\left(\frac{2\pi x}{a}\right) \\ &= -\frac{9\hbar}{5a} e^{-iRt} \int_0^\pi dy \sin(2y) \cos(3y) + \frac{6\hbar}{5a} e^{+iRt} \int_0^\pi dy \sin(3y) \cos(2y) \end{aligned}$$

with the substitution $y = \pi x/a$ in the last term. The necessary integrals are

$$\int_0^\pi dy \sin(2y) \cos(3y) = -\frac{4}{5} \quad , \quad \int_0^\pi dy \sin(3y) \cos(2y) = \frac{6}{5}$$

Using these integrals,

$$\begin{aligned} \langle \hat{p} \rangle &= \frac{36\hbar}{25a} (e^{-iRt} + e^{+iRt}) = \frac{72\hbar}{25a} \cos(Rt) \\ \Rightarrow \quad \langle \hat{p} \rangle &= \frac{72\hbar}{25a} \cos[(\omega_3 - \omega_2)t] = \frac{72\hbar}{25a} \cos\left[\frac{5\pi^2\hbar t}{2ma^2}\right] \end{aligned}$$

The momentum expectation value oscillates around zero. The oscillation amplitude has the correct dimensions of momentum: proportional to $\hbar/a$.

- (c) Use your results to confirm that $\langle \hat{p} \rangle = m \frac{d}{dt} \langle \hat{x} \rangle$

Hint/Solution $\rightarrow$

We obtained

$$\begin{aligned} \langle \hat{x} \rangle &= \frac{a}{2} + \frac{144a}{125\pi^2} \sin\left[\frac{5\pi^2\hbar t}{2ma^2}\right] \\ \langle \hat{p} \rangle &= \frac{72\hbar}{25a} \cos\left[\frac{5\pi^2\hbar t}{2ma^2}\right] \end{aligned}$$

Taking the derivative of the first equation w.r.t. time immediately gives $\langle \hat{p} \rangle = m \frac{d}{dt} \langle \hat{x} \rangle$.